OAVS • TGT COMPUTER SCIENCE

Network Database

Introduction | MCQ Question Bank

CHAPTER 4 · TOPIC 4.1

Introduction to Network Database

57	10	A–D
TOTAL MCQS	SECTIONS	BALANCED KEYS

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About this Question Bank

- Strictly limited to Topic **4.1 — Introduction to Network Database** as per the OAVS TGT syllabus.
- Contains **57 unique MCQs** covering definitions, concepts, structure, records, owner/member/set concepts, characteristics, advantages, limitations, applications, diagrams, and tricky conceptual questions.
- Difficulty mix follows the OAVS TGT pattern: approximately 25% Easy, 50% Moderate, 25% Difficult/Tricky.
- Answer keys are distributed in a balanced, non-predictable manner across options A, B, C and D (Distribution — A: 15 B: 14 C: 14 D: 14).
- Every question includes a concise explanation for quick revision.

Section 1 — Basic Concept of Network Database

Q1 DEFINITION NETWORK DATABASE

What is a Network Database?

A. A database that stores data only in tables

B. A database that works only over a computer network

C. A database model that represents data as a collection of records connected by links, allowing many-to-many relationships

D. A database used exclusively for internet applications

Ans: C

Explanation: The network database model represents data as interconnected records joined by links, which naturally supports many-to-many relationships.

Q2 TERMINOLOGY MEANING

The term "network" in Network Database primarily refers to

A. Internet connectivity between computers

B. Local Area Network configuration

C. Networking hardware used to store data

D. Interconnection of data records through links

Ans: D

Explanation: In this context, "network" describes the interconnected structure of data records, not physical computer networking hardware.

Q3 CONCEPT PURPOSE

What was the network database model primarily designed to overcome?

A. The high cost of relational databases

B. The limitation of the hierarchical model in representing many-to-many relationships

C. The complexity of SQL queries

D. The need for internet-based storage

Ans: B

Explanation: The hierarchical model could only represent one-to-many relationships; the network model was designed to handle many-to-many relationships.

Q4 CONCEPT BASIC STRUCTURE

Which of the following best fits the basic concept of a network database?

A. Data organized as records interconnected via multiple links forming a network

B. Data organized as isolated tables with no links

C. Data organized as a strict single-parent tree only

D. Data organized only as key-value pairs

Ans: A

Explanation: A network database's core idea is records connected through multiple links, forming a network-like structure rather than a strict tree.

Q5 TERMINOLOGY MODEL ORIGIN

The Network Database Model is closely associated with which standard specification?

A. Relational Model

B. CODASYL Model

C. Object Model

D. Entity Model

Ans: B

Explanation: The network model is commonly associated with the CODASYL (Conference on Data Systems Languages) DBTG specifications.

Q6 TERMINOLOGY BASIC UNIT

In a network database, the fundamental unit that represents a real-world entity is called a

A. Table

B. Tuple

C. Attribute list

D. Record

Ans: D

Explanation: A record is the basic unit of storage in a network database, representing a single entity through its data items.

Section 2 — Network Database Structure

Q7 STRUCTURE DATA ARRANGEMENT

In a network database structure, data is arranged as

A. A single flat file

B. A strict linear list

C. A graph-like structure of interconnected records

D. A single-level hierarchy only

Ans: C

Explanation: Network databases arrange records in a graph-like structure, allowing more complex connections than a simple list or hierarchy.

Q8 COMPARISON HIERARCHICAL VS NETWORK

Compared to the hierarchical model, the network model allows

A. A record to have multiple parent (owner) records

B. Only one parent per record

C. No relationships between records

D. Only tabular relationships

Ans: A

Explanation: Unlike the hierarchical model, the network model allows a record to participate as a member in more than one set, enabling multiple parents.

Q9 STRUCTURE VISUALIZATION

The overall structure of a network database can best be visualised as

A. A web/graph of interconnected nodes

B. A tree with a single root

C. A single continuous list

D. A grid of unrelated cells

Ans: A

Explanation: Because records can link to multiple other records, the structure resembles a web or graph rather than a simple tree.

Q10 STATEMENT RELATIONSHIP SUPPORT

Which statement about network database structure is TRUE?

A. It supports only one-to-one relationships

B. It supports only one-to-many relationships

C. It supports many-to-many relationships directly through sets

D. It cannot represent any relationships

Ans: C

Explanation: A key strength of the network structure is its direct support for many-to-many relationships through the set concept.

Q11 ASSERTION-REASON FLEXIBILITY

Assertion (A): Network database structure is more flexible than hierarchical structure.
Reason (R): Network structure allows a record to be linked to more than one owner record.

A. Both A and R are true, but R is not the correct explanation of A

B. Both A and R are true, and R is the correct explanation of A

C. A is true, R is false

D. A is false, R is true

Ans: B

Explanation: Both statements are true, and the ability to link to multiple owners is exactly why the network structure is more flexible.

Q12 SCENARIO MULTIPLE PARENTS

A student record is linked to both "Course" and "Hostel" record types at the same time. This structure is characteristic of

A. Hierarchical database

B. Flat file database

C. None of the above

D. Network database

Ans: D

Explanation: Allowing a single record to be linked to more than one owner record type is a defining trait of the network database model.

Section 3 — Records, Data Items and Relationships

Q13 TERMINOLOGY DATA ITEM

In network database terminology, a "data item" refers to

A. A complete record

B. A group of unrelated records

C. A type of link

D. The smallest unit of named data

Ans: D

Explanation: A data item is the smallest named unit of data; a group of related data items together forms a record.

Q14 TERMINOLOGY RECORD TYPE

A "record type" in a network database defines

A. A single link between two records

B. The graphical layout of the database

C. A group of data items describing a particular entity

D. Only the primary key of a table

Ans: C

Explanation: A record type is a template describing the set of data items that together represent a particular kind of entity.

Q15 CONCEPT RELATIONSHIPS

Relationships between records in a network database are established through

A. Links (pointers) connecting record occurrences

B. Foreign keys only

C. SQL joins

D. Indexes only

Ans: A

Explanation: Network databases connect related record occurrences directly using links (pointers), not foreign keys or SQL joins.

Q16 STATEMENT MULTIPLE RELATIONSHIPS

Which of the following is TRUE about relationships in a network model?

A. A record can participate in only one relationship

B. A record occurrence can participate in multiple relationships simultaneously

C. Relationships exist only between identical record types

D. Relationships cannot connect more than two records

Ans: B

Explanation: A record occurrence in a network database can take part in several relationships (sets) at the same time.

Q17 COMPARISON DATA ITEM ANALOGY

In a network database, a 'data item' is analogous to which relational database concept?

A. A table

B. A field/attribute

C. A row

D. A database

Ans: B

Explanation: A data item corresponds to a single field or attribute — the smallest named unit describing part of a record.

Q18 STATEMENT WORKING PRINCIPLE

Consider the statement: 'Each record occurrence in a network database can have links to multiple other records, forming complex relationships.' This statement is

A. False, because records can only link to one other record

B. False, because links are not allowed in network databases

C. True, but only applicable to hierarchical databases

D. True, and reflects the basic working principle of network databases

Ans: D

Explanation: This statement is correct and describes the fundamental navigational, multi-link nature of network databases.

Section 4 — Owner, Member and Set Concepts

Q19 TERMINOLOGY SET

In network database terminology, a "Set" represents

A. A collection of unrelated tables

B. A group of primary keys

C. A relationship between one owner record type and one or more member record types

D. A single isolated record

Ans: C

Explanation: A "set" is the named structure that defines a relationship between one owner record type and its member record type(s).

Q20 TERMINOLOGY OWNER

In an owner-member relationship, the "owner" record is

A. The parent-like record that "owns" one or more related member records

B. The record that depends on another record

C. A record with no links

D. Always the last record in the database

Ans: A

Explanation: The owner record acts like a parent in the relationship, controlling one or more associated member records.

Q21 TERMINOLOGY MEMBER

The "member" record in a network database set is best described as

A. The record that participates in a relationship as the dependent/related record

B. The record that owns other records

C. A record with no relationship to any other record

D. The topmost record in the structure

Ans: A

Explanation: The member record is the dependent record within a set, associated with its owner record.

Q22 CONCEPT MULTIPLE PARENTS

Which of the following correctly describes multiple parent relationships in network databases?

A. A member record can belong to only one owner

B. Owner records cannot have multiple members

C. Multiple parent relationships are not possible in network databases

D. A member record can be linked to multiple owner records through different sets

Ans: D

Explanation: A member record type can participate in several different sets, giving it links to multiple owner record types.

Q23 MATCH THE FOLLOWING OWNER/MEMBER/SET/RECORD

Match Column A with Column B.
Column A: (i) Owner (ii) Member (iii) Set (iv) Record
Column B: (a) Basic unit of data (b) Dependent record in a relationship (c) Relationship structure between records (d) Record that controls related records

A. i-a, ii-b, iii-c, iv-d

B. i-d, ii-b, iii-c, iv-a

C. i-b, ii-d, iii-a, iv-c

D. i-c, ii-a, iii-d, iv-b

Ans: B

Explanation: Owner = record that controls related records (d); Member = dependent record (b); Set = relationship structure (c); Record = basic unit of data (a).

Q24 COMPARISON OWNER VS MEMBER

A key difference between owner and member records is that

A. Owner records always have fewer data items than member records

B. Member records cannot have data items

C. An owner record can be associated with many member record occurrences, but a member typically belongs to fewer owners in a simple set

D. Owner and member are identical concepts

Ans: C

Explanation: In a basic set, one owner occurrence relates to many member occurrences — the defining owner-member relationship.

Section 5 — Network Characteristics

Q25 FEATURE BASIC CHARACTERISTIC

Which is a basic characteristic of network databases?

A. Strictly hierarchical single-parent structure

B. Absence of any relationship between records

C. Support for many-to-many relationships via sets

D. Data stored without any links

Ans: C

Explanation: Supporting many-to-many relationships through sets is a defining characteristic of network databases.

Q26 CONCEPT NAVIGATION

Navigation through a network database is typically done by

A. Random access without any connection

B. Following links between owner and member records

C. Using SQL SELECT statements only

D. Sorting all records alphabetically

Ans: B

Explanation: Network databases use a navigational approach, traversing links between owner and member records to reach data.

Q27 COMPARISON VS FILE SYSTEMS

Which characteristic distinguishes network databases from simple file systems?

A. Network databases store data without organization

B. File systems always use interconnected records

C. There is no difference

D. Network databases establish explicit links/relationships between records

Ans: D

Explanation: Explicit links/relationships between records set network databases apart from simple, unlinked file systems.

Q28 FEATURE NAVIGATIONAL ACCESS

A defining feature of the network model's navigational approach is that

A. Data can be accessed by traversing links between owner and member records in various paths

B. Data can only be accessed sequentially from the first record

C. Data cannot be accessed once stored

D. Data access requires converting to hierarchical format

Ans: A

Explanation: Navigational access means data is reached by following links along various possible paths, not a single fixed path.

Q29 STATEMENT NON-CHARACTERISTIC

Which of the following is NOT a characteristic of network databases?

A. Interconnected record structure

B. Support for multiple owner-member relationships

C. Navigational data access

D. Complete absence of relationships between records

Ans: D

Explanation: Network databases are defined by their relationships between records, so an absence of relationships contradicts the model entirely.

Section 6 — Advantages and Limitations

Q30 ADVANTAGE BASIC ADVANTAGE

A basic advantage of the network database model is

A. It efficiently represents many-to-many relationships between records

B. It cannot represent complex relationships

C. It requires no links between records

D. It is simpler than all other database models

Ans: A

Explanation: Efficient representation of many-to-many relationships is one of the main advantages of the network model.

Q31 LIMITATION BASIC LIMITATION

A basic limitation of the network database model is

A. It is too simple to represent any relationships

B. It cannot store any data items

C. Its complex structure makes navigation and maintenance difficult

D. It has no concept of records

Ans: C

Explanation: The complexity of the interconnected structure makes navigation and database maintenance challenging.

Q32 ADVANTAGE VS HIERARCHICAL

Which is an advantage of the network database over the hierarchical database?

A. Simpler structure with fewer relationships

B. Ability to represent multiple parent-child (many-to-many) relationships

C. No need for links between records

D. Faster because it avoids using records

Ans: B

Explanation: The network model's ability to represent multiple-parent, many-to-many relationships is its key advantage over the hierarchical model.

Q33 LIMITATION USABILITY

A key limitation of network databases regarding usability is that

A. They require complex navigational programming to access data

B. They cannot be used for any real application

C. They do not support records

D. They eliminate the need for data items

Ans: A

Explanation: Accessing data typically requires writing navigational code to traverse links, which is more complex than declarative querying.

Q34 ASSERTION-REASON COMPLEXITY

Assertion (A): Network databases are considered complex for application developers.
Reason (R): The network database model was developed before the relational model became popular.

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true, but R is not the correct explanation of A

C. A is true, R is false

D. A is false, R is true

Ans: B

Explanation: Both statements are factually true, but the timing of development does not explain why the model is complex — the complexity comes from navigational access, not chronology.

Q35 ADVANTAGE STRENGTH

Which of these is typically considered a strength of the network model?

A. Poor representation of relationships

B. Inability to link records

C. Flexibility in modeling complex, interconnected data

D. Restriction to one-to-one relationships only

Ans: C

Explanation: The network model's flexibility in modeling complex, interconnected data relationships is one of its key strengths.

Section 7 — Applications and Examples

Q36 APPLICATION USE CASE

Network databases are historically well-suited for applications requiring

A. Simple flat data with no relationships

B. Only textual document storage

C. Single-user spreadsheet data

D. Complex interconnected data such as engineering and manufacturing systems

Ans: D

Explanation: Applications with complex interconnected data, such as engineering and manufacturing systems, suit the network model well.

Q37 EXAMPLE IMPLEMENTATION

Which is a commonly cited early example/implementation associated with the network database model?

A. Oracle

B. IDMS (Integrated Database Management System)

C. MySQL

D. MongoDB

Ans: B

Explanation: IDMS is a well-known example of a network database management system based on the CODASYL specifications.

Q38 CONCEPT DATABASE EVOLUTION

Network database concepts are foundational to understanding

A. The evolution of database models leading to relational databases

B. Only spreadsheet software

C. Word processing applications

D. Operating system design

Ans: A

Explanation: Studying the network model helps explain how database design evolved toward the later relational model.

Q39 APPLICATION MANY-TO-MANY USE

In which type of application would many-to-many relationships (a key network database feature) be most useful?

A. A single list of names with no connections

B. A single-file text document

C. A calculator application

D. A student-course enrollment system where students take many courses and courses have many students

Ans: D

Explanation: Student-course enrollment is a classic many-to-many scenario, well suited to the network model's set structures.

Q40 APPLICATION HISTORICAL USE

Which of the following types of systems historically used network database concepts?

A. Static personal blogs

B. Basic text editors

C. Airline reservation and inventory management systems

D. Simple calculator programs

Ans: C

Explanation: Airline reservation and inventory management systems historically relied on network database structures for their complex, interlinked data.

Section 8 — Diagram-Based Questions

Q41 DIAGRAM OWNER IDENTIFICATION

Study the diagram below, where 'Department' is connected to 'Employee 1' and 'Employee 2'. Which record in the diagram is the owner?



A. Employee 1

B. Employee 2

C. Cannot be determined

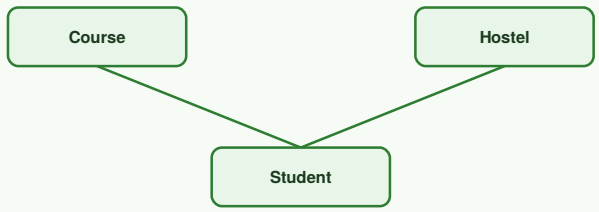
D. Department

Ans: D

Explanation: 'Department' controls the linked employee records, making it the owner record in this set.

Q42 DIAGRAM MULTIPLE OWNERS

Study the diagram below, where both 'Course' and 'Hostel' are connected to the same 'Student' record. What does this diagram illustrate?



A. A simple one-to-one relationship

B. No relationship at all

C. A multiple parent (many-to-many) relationship

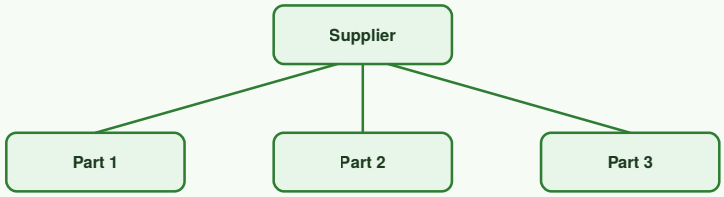
D. A hierarchical single-parent structure

Ans: C

Explanation: A single member record connected to two different owner records demonstrates the multiple-parent capability unique to network databases.

Q43 DIAGRAM LINKS AS SETS

Study the diagram below, where 'Supplier' is linked to 'Part 1', 'Part 2', and 'Part 3'. In this diagram, the links represent



A. Data items within a single record

B. Relationships (sets) connecting owner and member records

C. Independent, unrelated records

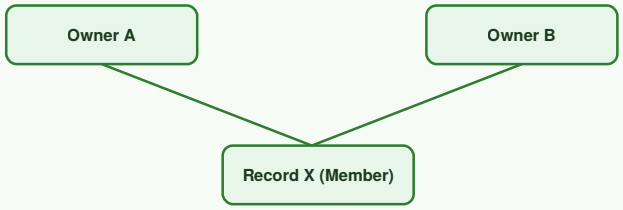
D. A hierarchical tree with no owner

Ans: B

Explanation: The lines in such diagrams represent sets — the relationships connecting an owner record to its member records.

Q44 DIAGRAM MEMBER WITH MULTIPLE OWNERS

Study the diagram below, showing a record with incoming links from two different owner records. What does this diagram best illustrate?



A. A member record participating in relationships with multiple owners

B. A record with no relationships

C. A record that owns itself

D. A broken link

Ans: A

Explanation: A record receiving links from two separate owners illustrates the network model's multiple-parent relationship feature.

Q45 DIAGRAM NAVIGATION PATH

Study the diagram below, showing an arrow-based path moving from an owner record to member records. This diagram represents which basic working principle of network databases?



A. Random data deletion

B. Navigational access through interconnected links

C. Table-based SQL querying

D. Data normalization

Ans: B

Explanation: Following a defined path along links from owner to member records is the essence of navigational access in network databases.

Section 9 — Conceptual and Tricky Questions

Q46 MISCONCEPTION COMMON CONFUSION

Which of the following is a common misconception about network databases?

A. That they use links to relate records

B. That they support many-to-many relationships

C. That they are the same as computer networks (LAN/WAN)

D. That records can have multiple owners

Ans: C

Explanation: A frequent misconception confuses the "network" database model with "network" as in computer networking (LAN/WAN); the two are unrelated concepts.

Q47 TRICKY MULTIPLE PARENT

If a record type participates as a member in two different sets with two different owner record types, this represents

A. A simple one-to-one link

B. An error in database design

C. A hierarchical structure

D. A multiple parent (multi-owner) relationship, a key network database feature

Ans: D

Explanation: Participating as a member in multiple sets with different owners is a valid and defining feature of network databases, not an error.

Q48 STATEMENT FALSE STATEMENT

Which statement is FALSE regarding network databases?

A. They restrict every record to exactly one parent, like hierarchical databases

B. They allow many-to-many relationships

C. They organize data using owner-member sets

D. They use links to connect records

Ans: A

Explanation: This statement is false — unlike hierarchical databases, network databases do not restrict a record to a single parent.

Q49 ASSERTION-REASON MODEL COMPARISON

Assertion (A): The network database model cannot represent relationships beyond one-to-one.
Reason (R): The network model allows a member record to have more than one owner record.

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true, but R is not the correct explanation of A

C. A is true, R is false

D. A is false, R is true

Ans: D

Explanation: Assertion A is false because the network model does represent many-to-many relationships; Reason R is a true statement describing this capability.

Q50 TERMINOLOGY TYPE VS OCCURRENCE

Which of the following correctly distinguishes 'record type' from 'record occurrence' in network database terminology?

A. Record type is a single instance of data; record occurrence is a template

B. Record type is the template/structure; record occurrence is an actual instance of that structure

C. Both terms mean exactly the same thing

D. Record type applies only to owner records

Ans: B

Explanation: A record type defines the structure (data items), while a record occurrence is one specific stored instance of that structure.

Q51 TRICKY SET MEANING

In network database terminology, a 'set' does NOT necessarily mean

A. A relationship between an owner and member record type

B. A named structure linking one owner type to member type(s)

C. A mathematical collection of unrelated numbers

D. A basic organizing concept for relationships

Ans: C

Explanation: In network database terminology, "set" refers to a named owner-member relationship structure, not a mathematical set of unrelated numbers — a common tricky distinction.

Section 10 — Mixed OAVS TGT Practice Questions

Q52 CONCEPT WORKING PRINCIPLE

Which of the following best summarises the basic working principle of a network database?

A. Data is organized as records connected through owner-member links enabling navigation across relationships

B. Data is stored in unrelated isolated files

C. Data is stored only as plain text

D. Data is accessed only through a single fixed path

Ans: A

Explanation: The network database's working principle centres on owner-member links that allow navigation across various relationships.

Q53 MATCH THE FOLLOWING MODEL CONCEPTS

Match Column A with Column B.
Column A: (i) Hierarchical limitation (ii) Network solution (iii) Set (iv) Navigation
Column B: (a) Owner-member relationship (b) Traversing links (c) Only one parent allowed (d) Represents many-to-many relationships

A. i-c, ii-d, iii-a, iv-b

B. i-a, ii-b, iii-c, iv-d

C. i-d, ii-c, iii-b, iv-a

D. i-b, ii-a, iii-d, iv-c

Ans: A

Explanation: Hierarchical limitation = only one parent allowed (c); Network solution = represents many-to-many relationships (d); Set = owner-member relationship (a); Navigation = traversing links (b).

Q54 STATEMENT MULTIPLE OWNER TYPES

Statement: 'In a network database, a member record type can be related to more than one owner record type.' This statement is

A. Incorrect, as members can only have one owner

B. Correct, but only applies to hierarchical databases

C. Correct, and reflects the multiple-parent capability of network databases

D. Incorrect, as network databases do not use owners

Ans: C

Explanation: This statement is correct — the multiple-parent capability is one of the defining features of the network database model.

Q55 COMPARISON MODEL MATCHING

Which of the following pairs is correctly matched?

A. Hierarchical model – supports many-to-many directly

B. Relational model – uses only network links

C. Network model – restricts records to one parent

D. Network model – organizes data via owner-member sets

Ans: D

Explanation: The network model is correctly characterised by its use of owner-member sets to organize data and relationships.

Q56 SCENARIO MULTI-OWNER EXAMPLE

A teacher explains that in a network database, 'Book' records can be linked to both 'Author' and 'Publisher' owner records simultaneously. This best demonstrates

A. A single hierarchical relationship

B. The many-to-many/multiple-owner capability of network databases

C. A database error

D. A flat file structure

Ans: B

Explanation: Linking one record to two different owner record types simultaneously demonstrates the network model's multiple-owner capability.

Q57 CONCEPT MODEL PURPOSE

Which best describes the overall purpose of introducing the network database model in database evolution?

A. To provide a more flexible model than hierarchical databases for representing complex, interconnected relationships

B. To eliminate the need for records

C. To replace the need for any data organization

D. To simplify data into single flat files only

Ans: A

Explanation: The network model was introduced to offer greater flexibility than the hierarchical model for representing complex, interconnected data relationships.